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EX NO: 8B

SIMULATION AND ANALYSIS OF QPSK MODULATION USING CONSTELLATION DIAGRAMS IN MATLAB

OBJECTIVES

- i) To generate QPSK symbols from a binary data sequence using quadrature phase-shift keying principles.
 - ii) To plot and analyze the QPSK signal constellation in the In-Phase (I) and Quadrature (Q) plane.
 - iii) To investigate the effect of noise on constellation points and evaluate the resulting degradation in symbol detection performance.
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PROGRAM (MATLAB CODE)

```
clc;
clear;
close all;
%% Objective 1: Generate QPSK Symbols
N = 1000;
data = randi([0 1], N, 1);
% Ensure even number of bits
if mod(N,2)~=0
    data = [data; 0];
    N = N + 1;
end
% Group bits into pairs
bit_pairs = reshape(data,2,[]);
% QPSK Mapping
qpsk_symbols = zeros(size(bit_pairs,1),1);
for k = 1:size(bit_pairs,1)
    if isequal(bit_pairs(k,:),[0 0])
```

```

    qpsk_symbols(k) = (1+1j)/sqrt(2);
elseif isequal(bit_pairs(k,:),[0 1])
    qpsk_symbols(k) = (-1+1j)/sqrt(2);
elseif isequal(bit_pairs(k,:),[1 1])
    qpsk_symbols(k) = (-1-1j)/sqrt(2);
elseif isequal(bit_pairs(k,:),[1 0])
    qpsk_symbols(k) = (1-1j)/sqrt(2);
end
end

%% Objective 2: Plot QPSK Constellation

figure;
scatter(real(qpsk_symbols),imag(qpsk_symbols),30,'filled');
hold on;
xline(0,'k','LineWidth',1.5);
yline(0,'k','LineWidth',1.5);
grid on;
axis equal;
xlim([-1 1]);
ylim([-1 1]);
xlabel('In-Phase (I)');
ylabel('Quadrature (Q)');
title('Ideal QPSK Constellation');

%% Objective 3: Add Noise and Evaluate Performance

SNR = 5;
rx_signal = awgn(qpsk_symbols,SNR,'measured');
figure;
scatter(real(rx_signal),imag(rx_signal),15,'filled');
hold on;
xline(0,'k','LineWidth',1.5);
yline(0,'k','LineWidth',1.5);
grid on;

```

```

axis equal;
xlabel('In-Phase (I)');
ylabel('Quadrature (Q)');
title(['Noisy QPSK Constellation (SNR = ',num2str(SNR),' dB)']);

%% Symbol Detection
detected_bits = zeros(length(rx_signal)*2,1);
for k = 1:length(rx_signal)
    I = real(rx_signal(k));
    Q = imag(rx_signal(k));
    if I>0 && Q>0
        bits = [0 0];
    elseif I<0 && Q>0
        bits = [0 1];
    elseif I<0 && Q<0
        bits = [1 1];
    else
        bits = [1 0];
    end
    detected_bits(2*k-1:2*k) = bits;
end

%% BER Calculation
num_errors = sum(data ~= detected_bits);
BER = num_errors/N;
fprintf('Number of Transmitted Bits : %d\n',N);
fprintf('Number of Bit Errors    : %d\n',num_errors);
fprintf('Bit Error Rate (BER)     : %f\n',BER);

```

PROCEDURE

Objective (i): Generate QPSK Symbols

1. Initialize the MATLAB environment and define the number of bits.
2. Generate a random binary sequence consisting of 0s and 1s.

3. Group the binary sequence into pairs of bits.
4. Apply QPSK modulation by mapping each bit pair to a unique constellation point.
5. Generate the corresponding complex QPSK symbols.
6. Verify that four distinct symbol values are obtained.
7. Store the generated symbols for constellation analysis.

Result:

QPSK symbols were successfully generated from the binary data sequence using quadrature phase-shift keying modulation.

Objective (ii): Plot and Analyze Constellation

1. Use the generated QPSK symbols for plotting.
2. Extract the real component as the In-Phase (I) coordinate.
3. Extract the imaginary component as the Quadrature (Q) coordinate.
4. Plot the constellation points on the I-Q plane.
5. Draw horizontal and vertical axes through the origin.
6. Identify the four symbol locations corresponding to bit pairs (00, 01, 11, 10).
7. Analyze distribution across the four quadrants.

Result:

The QPSK constellation was successfully plotted in the In-Phase (I) and Quadrature (Q) plane, showing four distinct symbol points corresponding to the bit pairs 00, 01, 11, and 10.

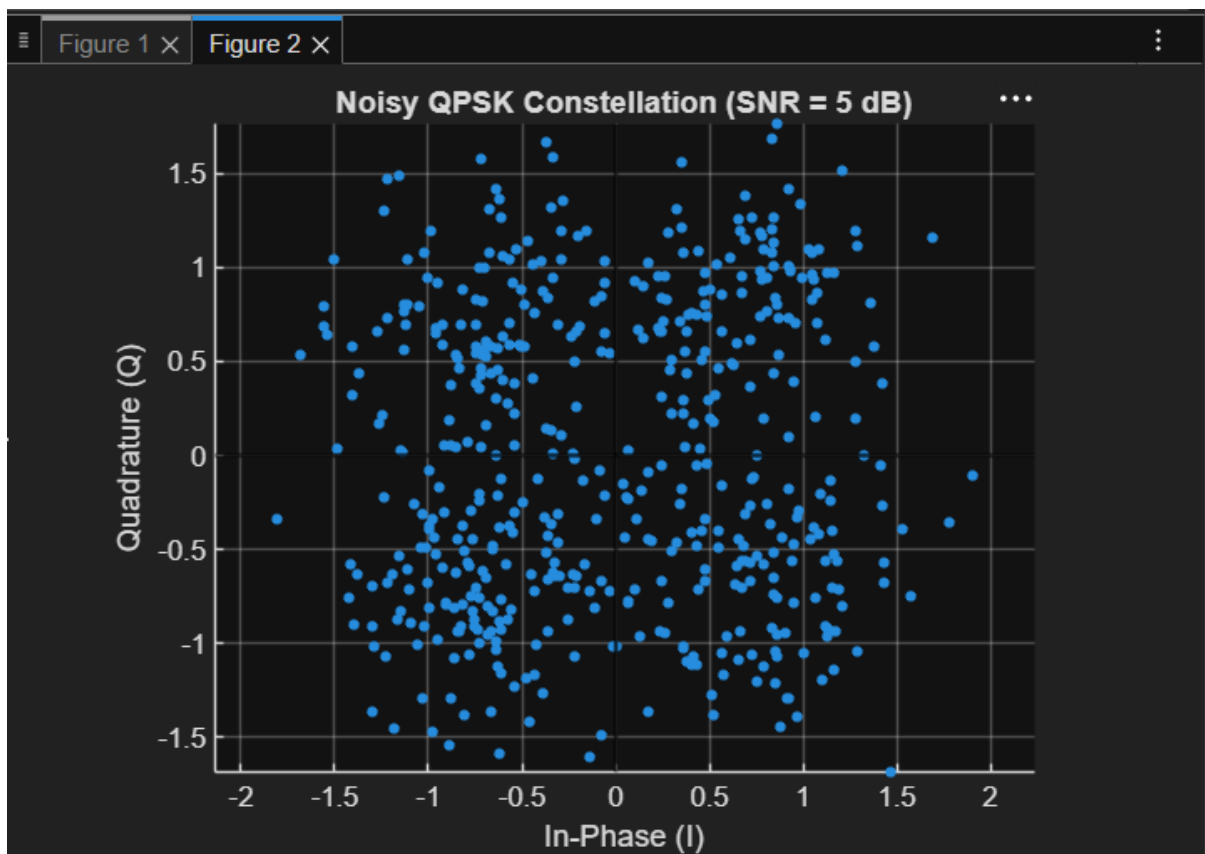
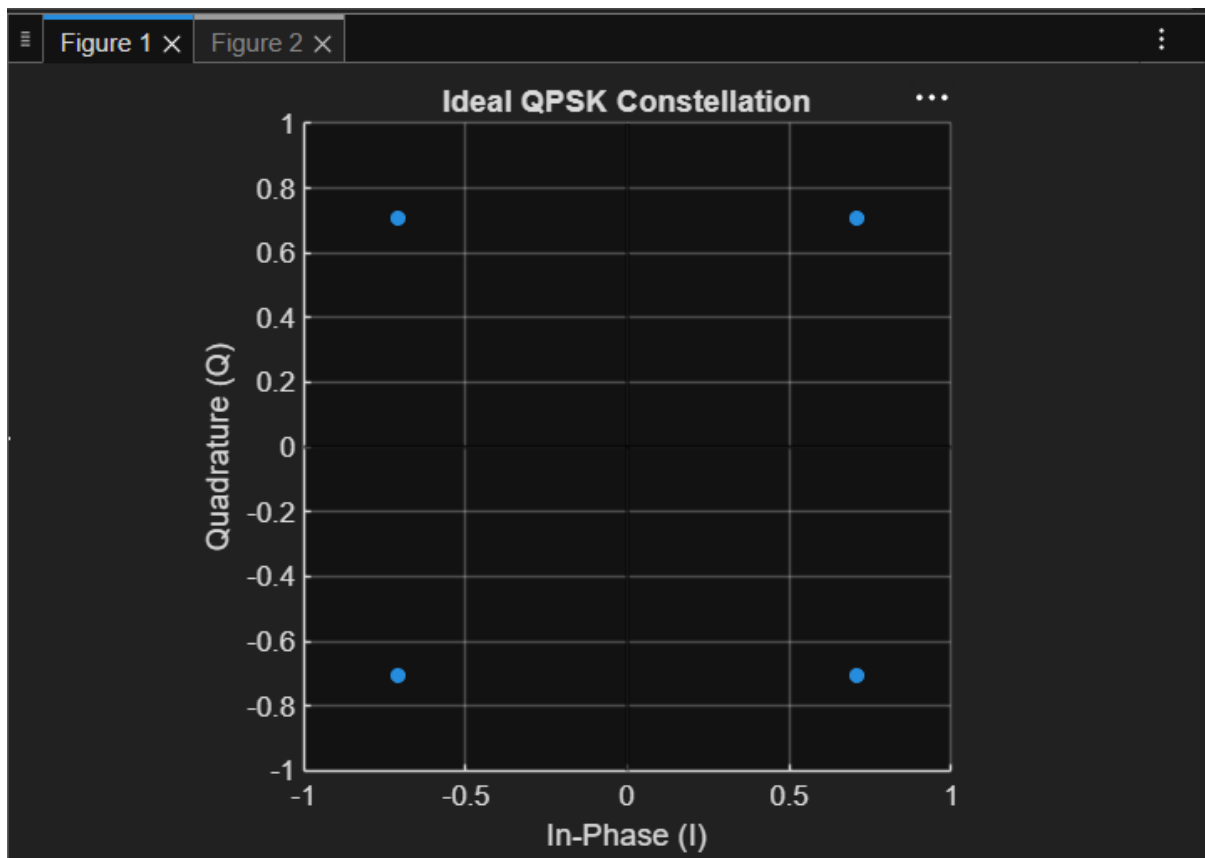
Objective (iii): Noise Effect and BER Analysis

1. Specify the Signal-to-Noise Ratio (SNR).
2. Add Additive White Gaussian Noise (AWGN) to the QPSK symbols.
3. Plot the noisy constellation diagram.
4. Observe spreading of constellation points around ideal positions.
5. Detect symbols using quadrant-based decision rules.
6. Recover transmitted bit pairs.
7. Compare transmitted and detected bits to calculate BER.

Result:

The effect of AWGN noise on the QPSK constellation points was successfully analyzed, and the degradation in symbol detection performance was evaluated using the Bit Error Rate (BER).

OUTPUT :



FINAL RESULT

The QPSK modulation was successfully simulated and analyzed using MATLAB. The constellation diagrams showed four distinct symbol points in ideal conditions, and the addition of noise caused dispersion, leading to bit errors. The system performance degradation was evaluated using BER.
